

The background of the slide is a dark blue gradient with various financial charts and data visualizations. In the top left, there are two line graphs with intersecting lines. To the right, there's a candlestick chart showing price movement. Below that, a line graph with several data points and connecting lines is visible. In the bottom right, there's a network diagram with nodes and connecting lines. The overall theme is financial data and analytics.

OptionQ

A Regime-Aware Options Hedge Engine

Dipen Prajapati

The Problem

Hedging is hard to get right — all at once.



Pick the wrong instrument → **basis risk**



Size incorrectly → **under/over hedged**



Ignore market conditions → **overpay in calm, underpay in stress**

Portfolio Input



Stocks



Options



12-Layer Pipeline



Ranked Hedge Candidates



Instrument +
Strike + Expiry



Contract
count



Market cost
(bid/ask)



Greeks per
position



Plain-English
explanation

Regime Detection

Main Model

10-year window,
4 components,
Retrain monthly

Fast Model

6-month window,
3 components,
Retrain daily

65%

35%

Combined Regime Signal

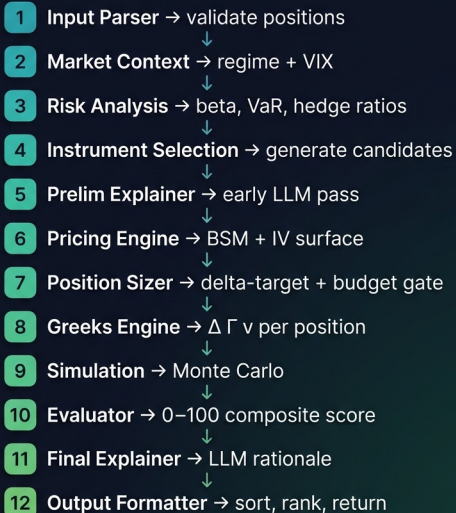
low_vol

mid_vol

high_vol

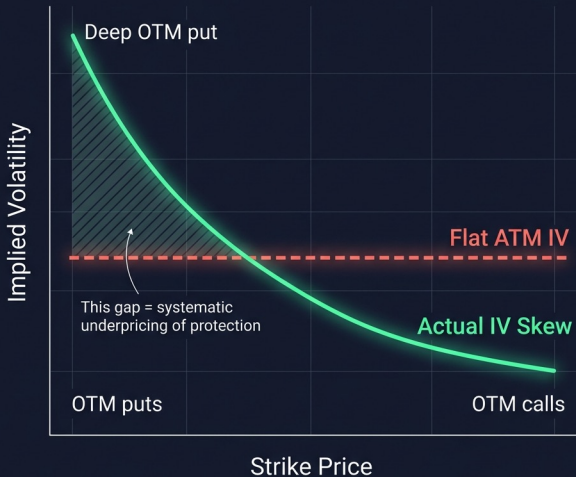
anomaly

The 12-Layer Pipeline



Premium

Accurate Pricing: IV Surface



Hedge Strategies + Real Cost

Strategy	Cost	Protection	Upside
Protective Put	High	Unlimited	Preserved
Bear Put Spread	Medium	Capped	Preserved
Collar	Near-zero	Moderate	Capped

Each candidate shows:

Market mid $(\text{bid} + \text{ask})/2$ ← what you pay at the broker,
BSM model price ← theoretical value

LLM Explainer — Layer 11

Sample AI-generated explanation

- ✓ Why this **hedge fits your position**
- ✓ **Pros** and **cons**
- ✓ When it **works best** / when it **fails**

Pluggable



Claude



Llama (local)

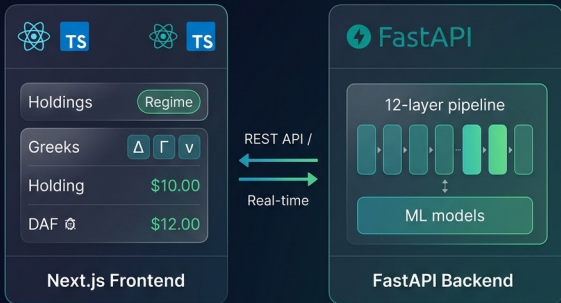


HuggingFace



Non-blocking: pipeline continues if LLM fails

Live UI — Architecture



Per holding:
top candidate's

Δ Γ ν

Per candidate:

- market cost
- model price
- coverage %

Regime badge:
current market
label

AI-generated
explanations

10210.00

StockOptionAAPL\$82,000

TypeDirection

CallPutLongShort

Strike (\$)Expiry

41006/16/2026

ContractsPremium paid (\$)

2optional

StockOptionMSFT\$4,680

SharesAvg price (\$)

12390.00

+ Stock+ Option

CONSTRAINTS\$88,780

Hedge horizon60d

Max cost3.5%

STOCK HEDGING

Protection level15%

Upside preservationOTM puts (keep upside)

MID VOLVIX 18.9Notional \$88,780Holdings 3Computed 1.95s

HEDGING YOUR POSITION

OptionAAPL Put \$410long · 2× · exp 2026-06-16Exposure \$82,000

2 candidatesBest score 53Direct Call (3% OTM) \$1,823 (2.2%)Δ +0.451Γ +2.6192v +88.21

153

Direct Call (3% OTM)Direct

AAPL CALL @ \$280 2026-06-21

\$1,8232.2% of pos.30% covereddelta +0.451

GREEKS — HOVER FOR EXPLANATION

Δ Delta+0.451

Γ Gamma+2.61924

Θ Theta-21.311

v Vega+88.208

Basis Risk R²1.00Low basis risk

Capital Eff. λ13.53×Protection / \$ premium

Max Protection\$24,66530% of position covered

RATIONALEDirect hedge on AAPL itself. R²=1.0 — zero basis risk. Calls offset short delta exposure. Strike 280 (exp 2026-06-21).

PROS

- Zero basis risk — tracks your position exactly
- No cross-asset correlation assumptions needed
- Expiry 2026-06-21 aligned to your position window

CONS

- Higher premium than ETF proxy (no diversification discount)
- Theta decay erodes value daily
- May have wider bid/ask spreads than index options
- Low liquidity: chain unavailable — verify before trading

247

Protective CallDirect

AAPL CALL @ \$275 2026-06-21

\$1,1311.4% of pos.17% covereddelta +0.517

HEDGING YOUR POSITION

StockMSFT12 shares · avg \$390Exposure \$4,680

POSITIONS

25

10

Stock

Option

AAPL

\$2,100

X

Shares

Avg price (\$)

10

210.00

Stock

Option

AAPL

\$82,000

X

Type

Direction

Call

Put

Long

Short

Strike (\$)

Expiry

410

06/16/2026

Contracts

Premium paid (\$)

2

optional

Stock

Option

MSFT

\$4,680

X

Shares

Avg price (\$)

12

390.00

+ Stock

+ Option

CONSTRAINTS

\$88,780

Hedge horizon

60d

30d

1yr

Max cost

3.5%

1%

10%

STOCK HEDGING

Protection level

15%

5%

30%

Upside preservation

OTM puts (keep upside)

Analyze Portfolio

MID VOL

VIX 18.9

Notional \$88,780

Hedges 3

Computed 1.95s

HEDGING YOUR POSITION

Option

AAPL Put \$410

long · 2* · exp 2026-06-16

Exposure \$82,000

2 candidates

Best score 53

Direct Call (3% OTM) \$1,823 (2.2%)

Δ +0.451

Γ +2.6192

ρ +85.21

1

53

Direct Call (3% OTM)

Direct

\$1,823

2.2% of pos.

30% covered

+0.451

delta

AAPL CALL @ \$200 2026-06-21

2

47

Protective Call

Direct

\$1,131

1.4% of pos.

17% covered

+0.517

delta

AAPL CALL @ \$275 2026-06-21

HEDGING YOUR POSITION

Stock

MSFT

12 shares · avg \$390

Exposure \$4,680

3 candidates

Best score 72

Direct Put (3% OTM) \$836 (17.9%)

Δ -0.339

Γ +0.9955

ρ +24.99

1

72

Direct Put (3% OTM)

Direct

\$848

net mid - \$80/contract

17.9% of pos.

314% covered

-0.339

delta

MSFT PUT @ \$420 2026-05-01

2

65

Protective Put

Direct

\$1,763

net mid - \$18/contract

37.1% of pos.

457% covered

-0.494

delta

MSFT PUT @ \$430 2026-05-15

3

65

Proxy Put (10% OTM)

Cross

\$408

net mid - \$4/contract

8.8% of pos.

184% covered

-0.112

delta

SPY PUT @ \$940 2026-06-18

HEDGING YOUR POSITION

Stock

AAPL

10 shares · avg \$210

Exposure \$2,100

3 candidates

Best score 75

Direct Put (3% OTM) \$435 (20.7%)

Δ -0.320

Γ +1.7599

ρ +24.98

1

75

Direct Put (3% OTM)

Direct

\$448

net mid - \$400/contract

28.7% of pos.

417% covered

-0.320

delta

AAPL PUT @ \$200 2026-05-15

2

71

Protective Put

Direct

\$858

net mid - \$80/contract

39.8% of pos.

670% covered

-0.515

delta

AAPL PUT @ \$275 2026-05-15

3

60

Proxy Put (15% OTM)

Cross

\$230

net mid - \$905/contract

18.1% of pos.

78% covered

-0.060

delta

SPY PUT @ \$905 2026-06-18

The background of the slide features a dark blue gradient with faint, semi-transparent financial charts. On the left, there is a candlestick chart showing price movement. Overlaid on this and other areas are several line graphs and curves. Two of these curves are labeled 'Delta' in a small, light blue font. Another curve on the right is labeled 'Gamma'. The overall aesthetic is technical and professional, typical of a financial or data-related presentation.

Thank You

Dipen Prajapati

 github.com/Dipen0210/OptionQ